Experiment and record

Follow the steps on the right to prepare three agar plates（瓊脂板）, incubate（培養） them together, then observe and measure.

This booklet provides fixed teaching-model images and raw diameter data for three independent plates. Complete the observations, means and conclusions yourself.

Your original research proposal has been saved separately. For classroom comparison, this activity will use 3 agar plates（瓊脂板）. You do not need to change your original design.

1 incubator（培養箱）: use the 30°C and 24-hour settings for the three plates together. X, Y, Z discs and a control: samples are carried on discs; the control has the same carrier alone.

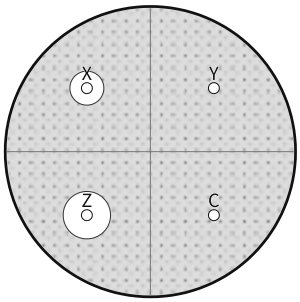


Plate 1

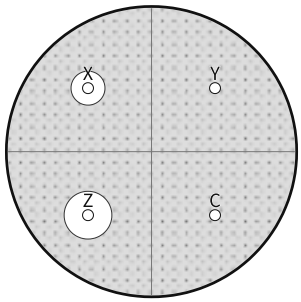


Plate 2

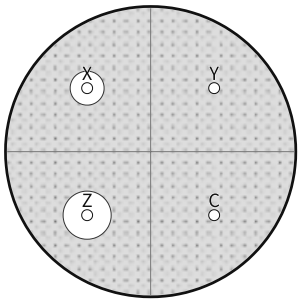


Plate 3

Measure and record the total clear-zone（清晰區） diameter (mm).

Plate	Sample	Raw diameter data (mm)	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
1	X	18.7	☐ Yes ☐ No	
	Y	0.0	☐ Yes ☐ No	
	Z	26.1	☐ Yes ☐ No	
	C (control)	0.0	☐ Yes ☐ No	
2	X	18.7	☐ Yes ☐ No	
	Y	0.0	☐ Yes ☐ No	
	Z	26.3	☐ Yes ☐ No	
	C (control)	0.0	☐ Yes ☐ No	
3	X	18.7	☐ Yes ☐ No	
	Y	0.0	☐ Yes ☐ No	
	Z	26.5	☐ Yes ☐ No	
	C (control)	0.0	☐ Yes ☐ No	

Images and raw diameters are generated by the current VL4 result model. No colony（菌落） counts, drug concentrations or patient doses have been added. Raw data are not mean or conclusion answers.

VL ruler guidance (paper uses the pre-rendered data above): Drag the ruler to measure; ticks are 1 mm. Near a clear-zone（清晰區） diameter it holds that height while remaining free to move left and right. Pull farther up or down to release.

Procedure

1. Flip the agar plate（瓊脂板）, divide its bottom into four areas with a cross, and label X, Y, Z and C.
2. Flip the agar plate（瓊脂板） back; use the dropper to collect diluted（已稀釋的） bacterial suspension（細菌懸液） and release it at the centre.
3. Move the sterile（無菌的） spreader（塗布器） up and down and rotate the agar plate（瓊脂板） to distribute the bacteria evenly.
4. Select the forceps and use the alcohol beaker followed by the lamp to sterilise（滅菌／消毒） them. Pick up a disc and place it at its labelled centre. Repeat sterilisation（滅菌／消毒） before picking up every disc.
5. Cover the agar plate（瓊脂板） and place it upside down.
6. Repeat steps 1–5 for agar plates（瓊脂板） 2 and 3.
7. When all three plates are ready, place them in the incubator（培養箱）, select 30°C and 24 hours, and start incubation（培養）.
8. For plates 2 and 3, you may choose to complete steps 1–5 automatically.

Revisions, if any (keep the first response readable; this is not an additional question)

Analyse and conclude

Use your observations and measurements to answer the multiple-choice questions and draw a conclusion.

Mean total diameter calculation

Plate	X (mm)	Y (mm)	Z (mm)	C (control) (mm)
1				
2				
3				

Mean for X (mm) _____ Mean for Y (mm) _____

Mean for Z (mm) _____ Mean for C (control) (mm) _____

1. What does the appearance of a clear zone（清晰區） indicate?

○ A. All bacteria in the zone have died.
○ B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
○ C. Bacteria in the zone resist every antibiotic（抗生素）.
○ D. The sample has eliminated viruses on the agar plate（瓊脂板）.

2. What does the diameter of a clear zone（清晰區） represent?

○ A. The exact number of bacteria remaining on the plate.
○ B. Its effectiveness in patients; the largest diameter identifies the best treatment.
○ C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
○ D. The antibiotic（抗生素） dose a patient should receive.

3. Draw a conclusion from your results:

Under these simulated conditions, （X ／ Y ／ Z ／ X and Y ／ X and Z ／ Y and Z ／ X, Y and Z ／ No samples） can effectively inhibit（抑制） MRSA（耐甲氧西林金黃葡萄球菌） growth.

Comparing the mean total clear-zone（清晰區） diameters in this test, their inhibitory（抑制） effects rank as follows:

（X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control)） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable） ＞ （X ／ Y ／ Z ／ C (control) ／ Not applicable）

Rank only samples with visible clear zones（清晰區）.

4. Why do experiments need to be repeated?

○ A. To ensure that every experiment produces exactly identical absolute values, with no variation whatsoever.
○ B. To prolong the experiment so scientists have more to describe when writing their reports.
○ C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
○ D. To make biological samples rapidly mutate and develop entirely new traits through repeated procedures.

5. If repeated results differ greatly, what should you do, and why?

○ A. Delete results that disagree with the original hypothesis so the results match.
○ B. Keep only the largest reading because it best represents inhibition（抑制）.
○ C. Calculate the mean immediately because averaging removes all procedural problems.
○ D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.

6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?

○ A. Ensure that all discs have clear zones（清晰區） of the same diameter.
○ B. Identify the best antibiotic（抗生素） for a patient.
○ C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.
○ D. Increase the amount of sample in the X, Y and Z discs to increase the accuracy（準確度） of the experimental results.

This model uses no clinical susceptibility（對抗生素的敏感性） breakpoints and cannot be generalised to all MRSA（耐甲氧西林金黃葡萄球菌）.

Revisions, if any (keep the first response readable; this is not an additional question)